Shivam Randive

Integrated M.Sc. Physics, SVNIT Surat, India | Machine learning for materials prediction

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Research Profile

My research focuses on discovering new alloys in the Heusler family and predicting their properties for thermoelectric applications, with machine-learning predictions tested against the ground truth they replace. One thread runs through my work. At the S. N. Bose Centre I showed that graph neural networks trained on narrow data fail on unseen materials. At SVNIT this grew into two first-author manuscripts with my supervisor, Dr. Himanshu Pandey: an ML screen of 106,235 Heusler structures judged by 3,625 first-principles calculations [P1], and a correction that makes ML predictions of thermal conductivity match experiment [P2]. At NIMS, Japan, I am now extending this to new alkali- and alkaline-earth-metal Heusler alloys, predicting their Seebeck coefficient and lattice thermal conductivity with transfer-learned interatomic potentials. **Interests:** graph neural networks, interatomic potentials, uncertainty-aware screening, generative design.

Education

Integrated M.Sc. (5-year BS–MS) in Physics — **2022 – 2027 (expected)**
Sardar Vallabhbhai National Institute of Technology (SVNIT) — *Surat, India*
CGPA **8.30 / 10** (top 20% of class)
Thesis: ML for materials discovery and property prediction in Heusler thermoelectrics

Semester Exchange Student — **Upcoming (Jan 2027)**
Indian Institute of Technology (IIT) Delhi — *New Delhi, India*

Publications

First-author manuscripts, both submitted to journals — * corresponding author

[P1] **S. Randive**, R. Dixit, H. Pandey. **High-throughput screening of Heusler alloys for thermoelectrics combining machine-learned densities of states and first-principles transport.** Submitted (2026). Preprint: arXiv:2610.18179.

[P2] **S. Randive***, O. Randive, H. Lungaria, H. Pandey*. **Predicting the experimental lattice thermal conductivity of half-Heusler compounds by machine learning and transfer-function calibration.** Submitted (2026); arXiv preprint forthcoming. Code & data: doi:10.5281/zenodo.23333947

Posters & Talks

Poster: *AI-driven discovery of Heusler alloys using generative models and electronic structure learning*, National Science Day, SVNIT Surat — Feb 2026

Research Internships

Research Intern · Spin Theory Group, Prof. Terumasa Tadano — **Sep – Dec 2026**
National Institute for Materials Science (NIMS) — *Tsukuba, Japan*

New alkali- and alkaline-earth Heusler thermoelectrics with pretrained AI models — ongoing

- *Idea:* build new alkali- and alkaline-earth Heusler compositions combinatorially and rank them by predicted Seebeck coefficient and lattice thermal conductivity, using universal machine-learned interatomic potentials (MLIPs: eSEN, EquiformerV3) pretrained on millions of DFT calculations.
- *Approach:* apply frozen transfer learning to these MLIPs on phonon force data, and predict lattice thermal conductivity through a Phonopy and Phono3py pipeline.

Summer Research Fellow (IASc SRFP) · Prof. Tanusri Saha-Dasgupta — **Jun – Jul 2025**
S. N. Bose National Centre for Basic Sciences — *Kolkata, India*

Do graph neural networks work on materials they have never seen? [report]

- Built a full ALIGNN pipeline and compared a broadly pretrained model with "specialists" trained only on narrow perovskite families.
- *Result:* specialists scored >92% on familiar data but only ≈48% on new double perovskites; the pretrained model kept ≈80%. *Message:* breadth of data beats narrow accuracy.

Summer Research Intern — **Jun – Jul 2024**
Indian Institute of Technology (IIT) Bhilai — *Bhilai, India*

Strain in silicon thin films for stretchable electronics [report]

- COMSOL finite-element model of a silicon film on flexible polyimide; its strain maps fed a collaborator's DFT study of how strain tunes the film's electronic properties.

Research Projects

Semester research projects in the Integrated M.Sc., Department of Physics, SVNIT, supervised by Dr. Himanshu Pandey

High-throughput ML screening of Heusler thermoelectrics with DFT validation [P1] 2026

- *Problem:* thermoelectrics turn waste heat into electricity. Searches for new ones first pick candidates from a cheap property, the density of states (DOS), often predicted by ML, before running costly calculations. No one had checked whether this shortcut picks the right materials.
- *What I did:* tested it on all 106,235 Heusler compounds in a large database. Graph neural networks predicted the DOS where none existed, and I checked the shortlist with full first-principles calculations run through my own automated workflow.
- *Result:* the shortcut is good at choosing what to calculate, needing about **4× fewer expensive calculations** per good material, and it found **80 promising candidates**, 61 never measured. But ML spectra often miss whether a material has a band gap, even when accuracy scores look excellent, so a full calculation must confirm it.

Calibrating ML predictions of lattice thermal conductivity against experiment [P2] 2026

- *Problem:* a good thermoelectric must conduct heat poorly. ML models that predict this learn from computer calculations, but I found that these calculations overestimate the measured value, typically by **about 70%**.
- *What I did:* gathered thermal-conductivity data from databases and published papers into one dataset that records where every value came from (AI models helped read the papers; every value was checked against physics), and built a simple correction that turns a calculated value into the one an experiment would measure.
- *Result:* with the correction, ML predicts measured values with **about 30% less error** than standard approaches, close to how much two laboratories disagree. I also released predictions for new compounds that experiments can test.

Simple physical descriptors as first predictors [report] Jan – Apr 2025

- Support-vector classifier for perovskite formability (≈90% accuracy); reproduced a Gaussian-process model of lattice constants for 137 half-Heusler compounds.

Next: Phono3py thermal conductivity for the [P1] candidates; crystal generation with MatterGen.

Open Data & Code

- **heusler-dos-te-screening** [P1] *(I designed and built it)*: the full screening pipeline (ML models, ranking tools, automated DFT transport workflow) with all 3,625 results and the 80 candidates.
- **Predicted DOS dataset** [P1] *(lead author; Zenodo, release pending)*: ML-predicted DOS for **54,978 Heusler structures** with no DFT spectrum, plus the 30 trained models.
- **half-heusler-kappa-calibration** [P2] *(lead author, archived on Zenodo)*: the 9,495-value thermal-conductivity dataset with the source of every value, the fits, the 10 predictions and the code that rebuilds every figure and number in the paper.

Fellowships & Awards

Fully funded research internship, National Institute for Materials Science (NIMS), Japan 2026
Selected for semester exchange at IIT Delhi; eligible for direct PhD admission 2026
Summer Research Fellowship (SRFP), Indian Academy of Sciences (national, merit-based) 2025
Best Structural Stability and Flight Award (3rd overall), national rocketry, IIT Mandi 2025
Xpecto tech fest; designed and launched a 250 g rocket with my team

Technical Skills

ML methods	Graph neural networks (training, fine-tuning, transfer learning); MLIP transfer learning and fine-tuning (eSEN, EquiformerV3); conformal prediction; SHAP
Classical ML	CatBoost, LightGBM, Gaussian processes, support-vector machines
ML libraries	PyTorch, DGL, fairchem, scikit-learn, NumPy, pandas, Matplotlib
First principles	VASP, Quantum ESPRESSO (incl. spin–orbit), BoltzTraP2, Phonopy/Phono3py
Materials data	Materials Project, AFLOW, JARVIS, OQMD, Starrydata; LLM literature mining
Computing	Python, Linux, SLURM GPU clusters, Git, MATLAB, COMSOL Multiphysics, LaTeX

Mentoring

Mentored two junior students, O. Randive and H. Lungaria; both are co-authors on [P2].